Oleksandr Nekrashevych

+27 66 558 3327 | shonnekrashevych@gmail.com
linkedin.com/in/oleksandr-nekrashevych | github.com/NekrashevychOleksandr | portfolio

Education

University Of Pretoria Expected December 2026
BEng Computer Engineering *Pretoria, Gauteng*

- **Relevant Coursework:** Control Systems, Sensor Fusion, State Estimation (KF, EKF, UKF), AI & Neural Networks, Computer Vision, Digital Signal Processing, Robotics, Embedded Systems, Electronics, Algorithms, Linear Algebra, Probability & Statistics

Projects

Facial Feature Extraction & CNC Sketching Robot | *Python, C, Embedded Systems, Computer Vision*

- Engineered an end-to-end computer vision and robotic sketching system as a final-year project, developed from first principles.
- Built an embedded image-processing pipeline on an Odroid N2+ using HOG-based descriptors, custom edge detection, and pixel-density feature localization to extract facial structures from images.
- Implemented an ESP32-controlled CNC drawing system that translated extracted facial features into motion commands, enabling automated sketch reconstruction on canvas with consistent geometric accuracy.

GAIA Autonomous Intelligence System | *Python, OpenCV, LLMs, Asynchronous Systems*

- Developed a modular autonomous AI agent framework with persistent memory, contextual reasoning, and emotional state tracking, currently supporting task execution through structured computer interaction workflows.
- Integrated real-time avatar expression via VTube Studio and implemented remote interaction systems through Cloudflare-hosted control interfaces and Telegram-based communication.
- Built an autonomous computer interaction pipeline using OpenCV screen analysis, Stockfish-based decision-making, and input automation to execute real-time chess gameplay without external APIs.

Autonomous Line-Following Robot (TUKS Robot Race Day – Semi-Finalist) | *Assembly, Embedded Systems*

- Designed and implemented an autonomous mobile robot capable of real-time line detection and trajectory following using onboard sensors and embedded control.
- Developed low-level assembly control firmware on the PIC18F45K22 microcontroller for direct motor actuation based on real-time sensor feedback, enabling precise navigation under dynamic conditions.
- Achieved semi-finalist placement in the TUKS Robot Race Day through consistent high-accuracy autonomous performance in variable track environments.

Experience

Cheery Robot Academy June 2024 – Present
Programming and Robotics Instructor *Pretoria, Gauteng*

- Taught programming and game development using Java, Python, C++, Lua, Roblox Studio, and Godot across school and university-level students.
- Delivered practical instruction in software development including automation, game development, and embedded systems fundamentals using Arduino-based robotics.
- Designed and supervised robotics and embedded systems projects involving microcontrollers and sensors to develop interactive hardware systems.
- Improved student performance across multiple cohorts, with several learners achieving distinction-level outcomes in programming modules.

Upwork July 2024 – June 2025
Freelance Software Developer *Remote*

- Developed 80+ SQL exercises for a Udemy database course used by 5,700+ learners, covering joins, indexing, constraints, subqueries, and query optimization.
- Provided technical input on PostgreSQL schema design and database structuring to support course development and learning progression.
- Worked directly with clients in a remote freelance environment, delivering production-ready educational content with consistent 5-star feedback.

RunJumpyFly June 2024 – July 2024

Frontend Development / Embedded Systems Intern *Pretoria, Gauteng*

- Developed a responsive web page for a financial consultancy website based on Figma design specifications, using Webflow, HTML, CSS, and JavaScript.
- Implemented reusable frontend components and interactive UI elements to improve layout consistency, navigation flow, and cross-device responsiveness.
- Supported creative embedded systems work by developing ESP32-based LED control logic for programmable lighting sequences.

KeyStone Electronic Solutions June 2022

Backend Engineering Intern *Pretoria, Gauteng*

- Worked on a legacy Java Spring Boot application and resolved dependency and library version conflicts that prevented execution on modern Linux environments.
- Integrated a MySQL database to enable persistent data storage and retrieval within the backend system.
- Containerised the application using Docker and aligned compatible dependency versions to ensure stable deployment on Ubuntu Linux systems.

Certifications

Cisco Networking Academy December 2020 – January 2024

Cisco Networking Certifications *Pretoria, Gauteng*

- CCNA: Introduction to Networks
- CCNA: Routing and Switching Essentials
- CCNAv7: Switching, Routing, and Wireless Essentials
- CCNAv7: Enterprise Networking, Security and Automation
- CCNP Enterprise: Core Networking

Technical Skills

Languages: Python, C, C++, Java, JavaScript, Lua, Assembly
Technologies: OpenCV, Docker, MySQL, PostgreSQL, MongoDB, Git, Embedded Systems, Hardware Interfacing, MATLAB, LTspice, Webflow, HTML, CSS
Concepts: Computer Vision, Embedded Systems, Control Systems, Robotics, Sensor Fusion, State Estimation, Kalman Filtering, Extended Kalman Filter, Real-Time Systems, Autonomous Systems, Agent Architectures, Asynchronous Programming, Image Processing, Signal Processing, Linux, Software Architecture